

Subject Name: Design of Machine Elements

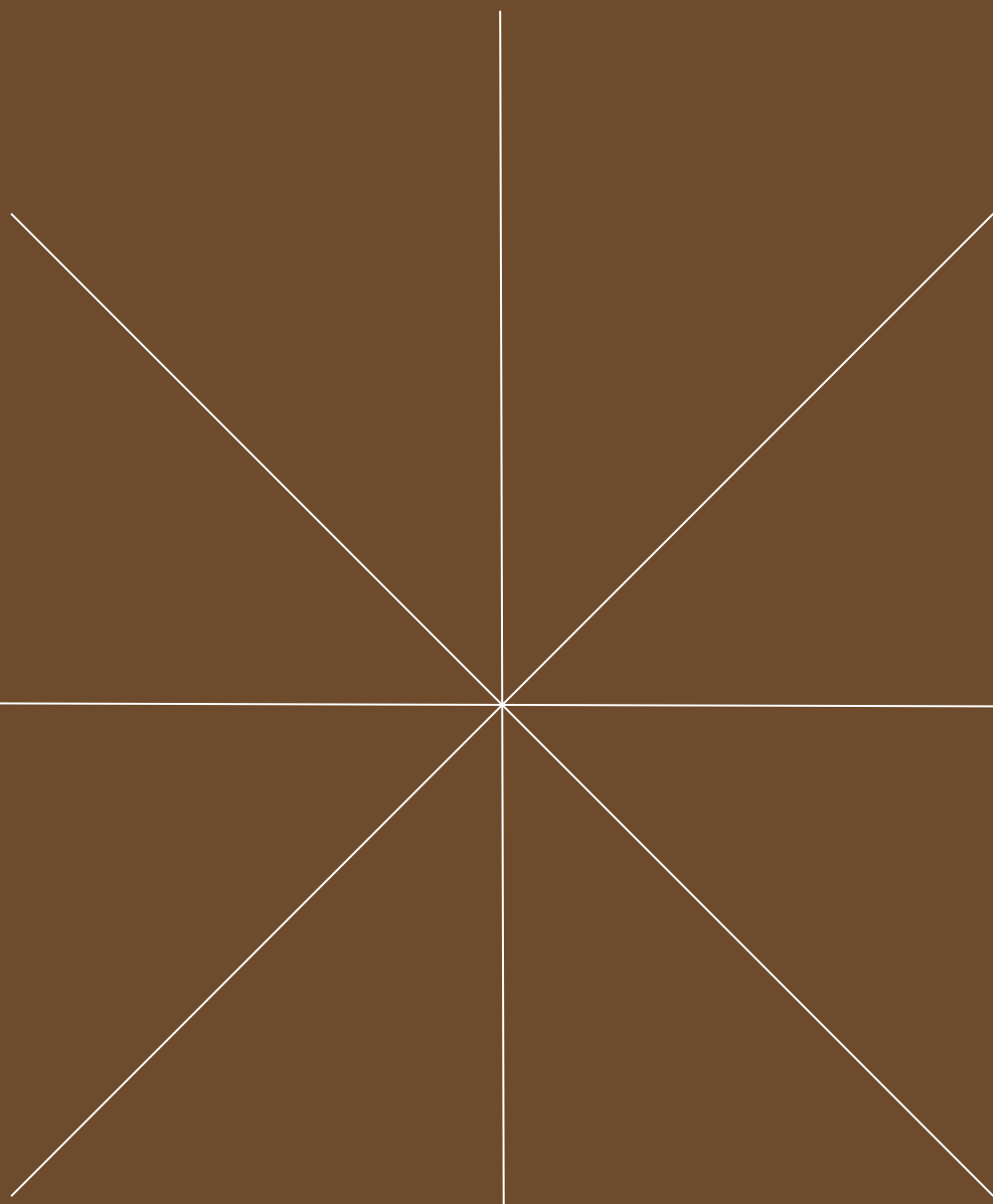
Subject Code: PC – ME 602

**Topic Name: Buckling analysis: Johnson and Euler
columns**

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Buckling Analysis

Johnson and Euler Columns Overview



Introduction

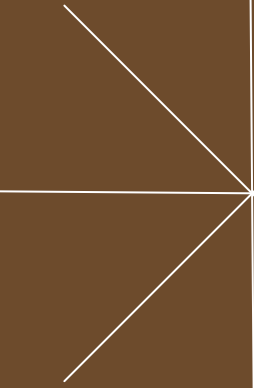
Buckling is a critical failure mode in slender structural members subjected to compressive loads. Understanding buckling behavior helps ensure safety and efficiency in design. This presentation covers the fundamentals of buckling with a focus on *Johnsons* and *Eulers* column theories, highlighting their assumptions, applications, and differences.

Definition and Importance of Buckling

Buckling is the sudden lateral deflection of a structural member under compressive stress, often causing collapse. It is crucial in design to prevent unexpected failures. The *critical buckling load* defines the maximum load before instability occurs, emphasizing the role of geometry, material properties, and boundary conditions. Proper buckling analysis ensures safety and optimization in structures.

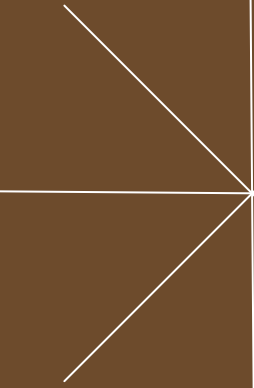


Types of Buckling in Structural Members



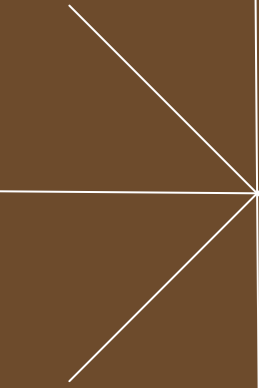
Buckling can occur in various forms: *flexural buckling* (bending in columns), *torsional buckling* (twisting), and *flexural-torsional buckling* (combination). The mode depends on member shape, supports, and load type. Each type influences design criteria differently, requiring tailored analysis to accurately predict failure points and choose appropriate preventive measures.

Critical Load Concept



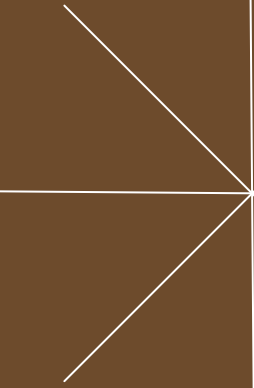
The *critical load* is the maximum axial load a column can bear before buckling occurs. It depends on factors like column length, cross-sectional area, material properties, and end conditions. Calculating this load ensures the column remains stable under compressive forces, preventing sudden failure and ensuring structural integrity throughout the service life of the component.

Euler's Column Theory: Assumptions and Applications



Euler's theory applies to long, slender columns experiencing *elastic buckling*. It assumes perfect straightness, homogeneous material, and axial load acting through the centroid. The critical load is derived by considering flexural rigidity and boundary conditions. Euler's formula is widely used in design where columns buckle elastically without material yielding.

Johnson's Parabolic Formula for Inelastic Buckling



Johnson's formula addresses *inelastic buckling* in short and intermediate columns where the material yields before buckling. It corrects Euler's theory by incorporating the yield strength and slenderness ratio. The parabolic nature provides safe estimates of critical load before plastic deformation, bridging the gap between elastic and plastic buckling behavior in design calculations.



Comparison between Euler and Johnson Approaches

Euler's theory suits *long, slender columns* exhibiting elastic behavior, predicting critical load accurately under ideal conditions. Johnson's formula is better for *shorter columns* where material yielding occurs before buckling. The choice depends on slenderness ratio and material limits, ensuring conservative and realistic designs by combining elastic and inelastic buckling considerations.



Conclusions

Buckling analysis is vital for structural safety, with Euler and Johnson models offering complementary approaches. Euler's theory governs elastic buckling in slender columns, while Johnson's formula handles inelastic buckling in shorter columns. Understanding their assumptions and application ranges allows engineers to select accurate methods, optimizing design to prevent failure and increase reliability.

Thanks!

